INORGANIC CUMULATIVE EXAMINATION

December 1, 1990

1. The visible spectrum of $Ti(H_2O)_6^{+3}$ shows one electronic transition, the spectrum of the corresponding Cr(III) complex shows at least two and under certain conditions three distinct transitions. In both cases an electron is promoted from a t_{2g} orbital to an e_g orbital. Why then does chromium show more than one electronic transition?

2. Elementary theory predicts a single electronic transition for the titanium(III) case presented above yet the absorption band shows a slight asymmetry indicative of two closely spaced transitions. What phenomena is responsible for this splitting? Explain. (Note: The answer to this question is not related to the answer to question one.)

3. The absorption spectrum of Mn(II) solutions shows five or more very sharp transitions. The exctinction coefficient for all of these transitions is almost two orders of magnitude smaller than those of the transitions discussed above. Why?

4. What is the Stark Effect?